COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31s2
DATE PERFORMED:	28/89/2026	**DATE SUBMITTED:**	28/08/2026
NAME:	Damien Jake D. Gestiada	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 5.1: **Implement Ansible roles in playbooks**

1. PROCEDURE OUTPUT

Task 1: Use when command for different distributions

1. In the local machine, make sure you are in the local repository directory (CPE232_yourname). Issue the command git pull. When prompted, enter the correct passphrase or password. Describe what happened when you issue this command. Did something happen? Why?

```
jakeworkstation@WS:~$ cd ~/ansible-lab
jakeworkstation@WS:~/ansible-lab$ ls
challenge.yaml   install_apache.yml  playbook.yaml  testvars.yaml
challenge.yaml]  inventory.yaml      README.md
jakeworkstation@WS:~/ansible-lab$ git pull
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (1/1), done.
remote: Total 2 (delta 1), reused 1 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (2/2), 879 bytes | 175.00 KiB/s, done.
From https://github.com/DamienJakeGestiada/CPE212_JakeGestiada
   c073d17..e3cc270  main       -> origin/main
Updating c073d17..e3cc270
Fast-forward
 challenge.yaml] | 1 -
 1 file changed, 1 deletion(-)
 delete mode 100644 challenge.yaml]
```

- The git pull command checks the GitHub repository for newer changes and updates the local repository. If it says "Already up to date," nothing needed to be downloaded because the local repository already has the latest version.

2. Edit the inventory file and add the IP address of the Centos VM. Issue the command we used to execute the playbook (the one we used in the last activity): ansible-playbook --ask-become-pass install_apache.yml. After executing this command, you may notice that it did not become successful in the Centos VM. You can see that the Centos VM has failed=1. Only the two remote servers have been changed. The reason is that Centos VM does not support "apt" as the package manager. The default package manager for Centos is "yum."

```
 GNU nano 7.2                                inventory.yaml
all:
  children:
    ubuntu:
      hosts:
        ubuntu1:
          ansible_host: 192.168.56.105
          ansible_user: jakeserver
          os_type: Ubuntu

        ubuntu2:
          ansible_host: 192.168.56.103
          ansible_user: jakeserver2
          os_type: Ubuntu

    centos:
      hosts:
        centos1:
          ansible_host: 192.168.56.104
          ansible_user: jakecentos
          os_type: CentOS

    linuxservers:
      children:
        ubuntu:
        centos:
```

```
jakeworkstation@WS:~/ansible-lab$ ansible all --list-hosts -i inventory.yaml
  hosts (3):
    ubuntu1
    ubuntu2
    centos1
jakeworkstation@WS:~/ansible-lab$
```

```
TASK [Update repository index] ************************************************************
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
fatal: [centos1]: FAILED! => {"changed": false, "cmd": "apt-get update", "msg": "[Errno 2] N
o such file or directory: b'apt-get'", "rc": 2, "stderr": "", "stderr_lines": [], "stdout":
"", "stdout_lines": []}
changed: [ubuntu1]
changed: [ubuntu2]

TASK [Install apache2 package] ************************************************************
ok: [ubuntu2]
ok: [ubuntu1]

TASK [Add PHP support for apache] *********************************************************
ok: [ubuntu2]
ok: [ubuntu1]

PLAY RECAP ********************************************************************************
centos1                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0
rescued=0    ignored=0
ubuntu1                    : ok=4    changed=1    unreachable=0    failed=0    skipped=0
rescued=0    ignored=0
ubuntu2                    : ok=4    changed=1    unreachable=0    failed=0    skipped=0
rescued=0    ignored=0

jakeworkstation@WS:~/ansible-lab$
```

3. Edit the install_apache.yml file and insert the lines shown below.

Make sure to save the file and exit.

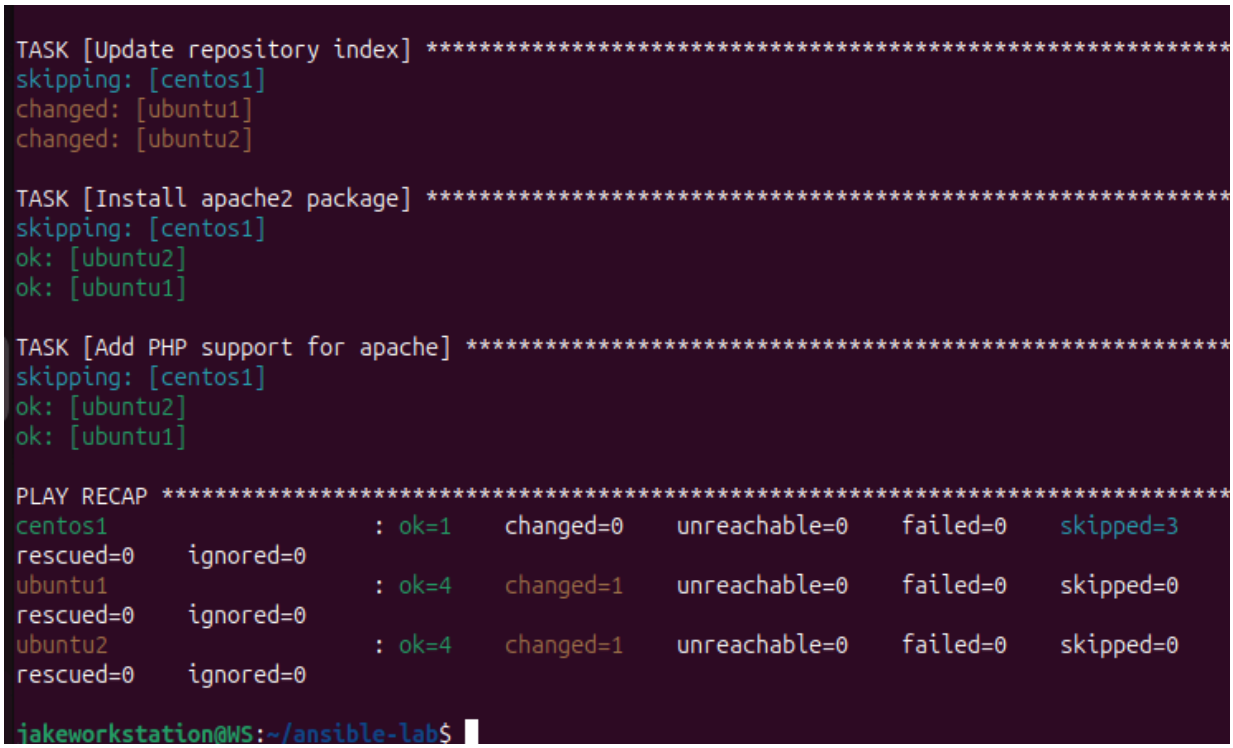
Run ansible-playbook --ask-become-pass install_apache.yml and describe the result.

If you have a mix of Debian and Ubuntu servers, you can change the configuration of your playbook like this.

- name: update repository index
apt:
update_cache: yes
when: ansible_distribution in [“Debian”, “Ubuntu]

Note: This will work also if you try. Notice the changes are highlighted.



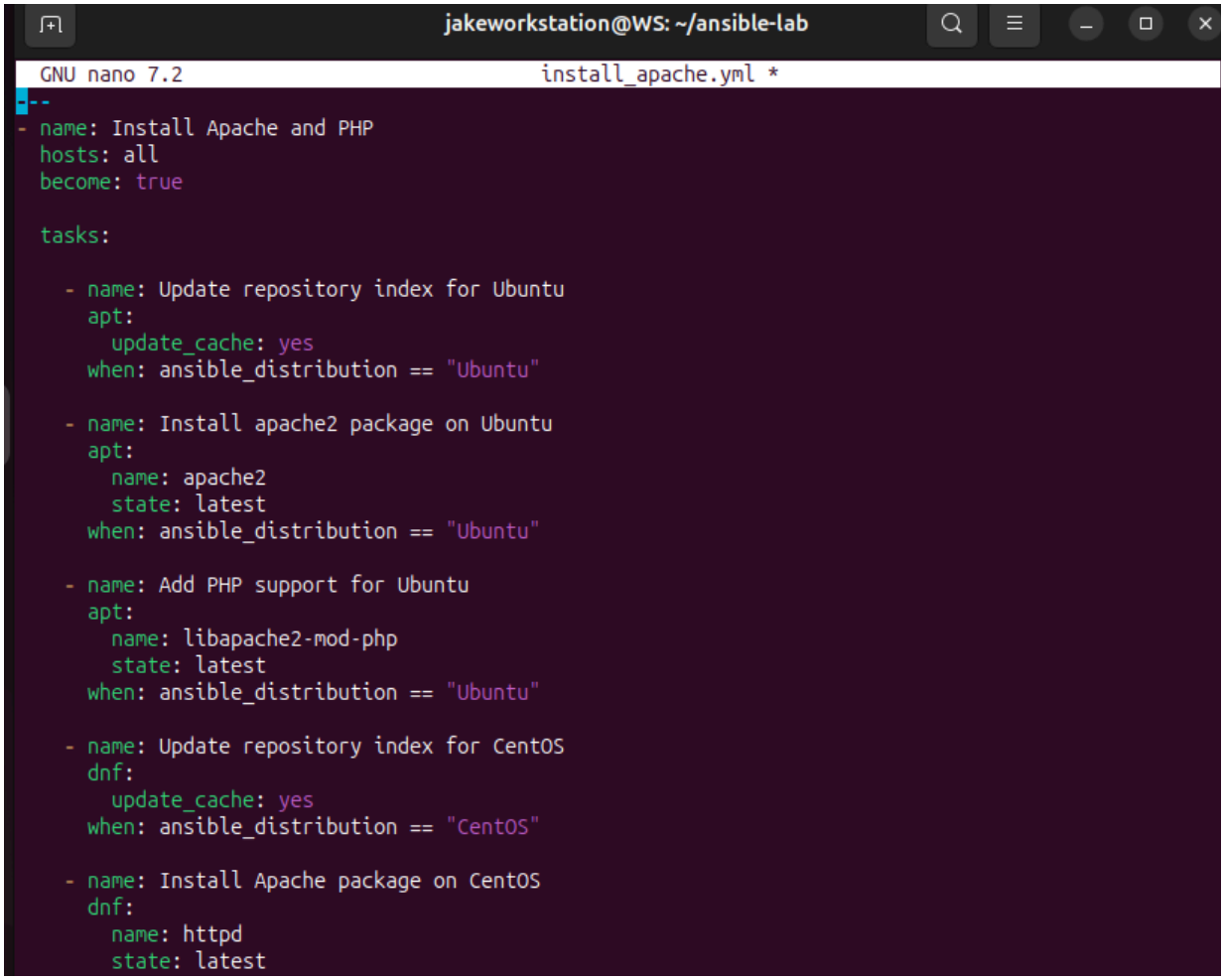


4. Edit the install_apache.yml file and insert the lines shown below.

Make sure to save and exit.

Run ansible-playbook --ask-become-pass install_apache.yml and describe the

Result.


jakeworkstation@WS: ~/ansible-lab
GNU nano 7.2
install_apache.yml *

- name: Install Apache and PHP
hosts: all
become: true
tasks:
- name: Update repository index for Ubuntu
apt:
update_cache: yes
when: ansible_distribution == "Ubuntu"
- name: Install apache2 package on Ubuntu
apt:
name: apache2
state: latest
when: ansible_distribution == "Ubuntu"
- name: Add PHP support for Ubuntu
apt:
name: libapache2-mod-php
state: latest
when: ansible_distribution == "Ubuntu"
- name: Update repository index for CentOS
dnf:
update_cache: yes
when: ansible_distribution == "CentOS"
- name: Install Apache package on CentOS
dnf:
name: httpd
state: latest

```
changed: [ubuntu2]
changed: [ubuntu1]

TASK [Install apache2 package on Ubuntu] ***********************************************************
skipping: [centos1]
ok: [ubuntu2]
ok: [ubuntu1]

TASK [Add PHP support for Ubuntu] ******************************************************************
skipping: [centos1]
ok: [ubuntu2]
ok: [ubuntu1]

TASK [Update repository index for CentOS] **********************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

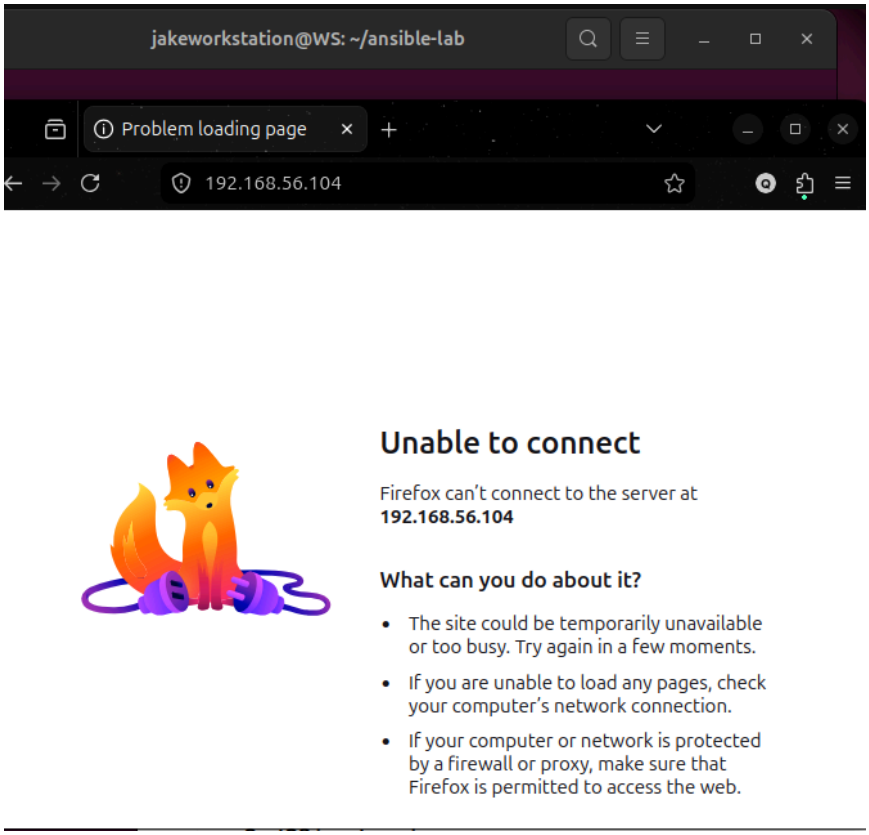
TASK [Install Apache package on CentOS] ************************************************************
skipping: [ubuntu2]
skipping: [ubuntu1]
ok: [centos1]

TASK [Add PHP support for CentOS] ******************************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

PLAY RECAP *****************************************************************************************
centos1                    : ok=4    changed=0    unreachable=0    failed=0    skipped=3    rescued=0
    ignored=0
ubuntu1                    : ok=4    changed=1    unreachable=0    failed=0    skipped=3    rescued=0
    ignored=0
ubuntu2                    : ok=4    changed=1    unreachable=0    failed=0    skipped=3    rescued=0
    ignored=0

jakeworkstation@WS:~/ansible-lab$
```

5. To verify the installations, go to CentOS VM and type its IP address on the browser. Was it successful? The answer is no. It's because the httpd service or the Apache HTTP server in the CentOS is not yet active. Thus, you need to activate it first.

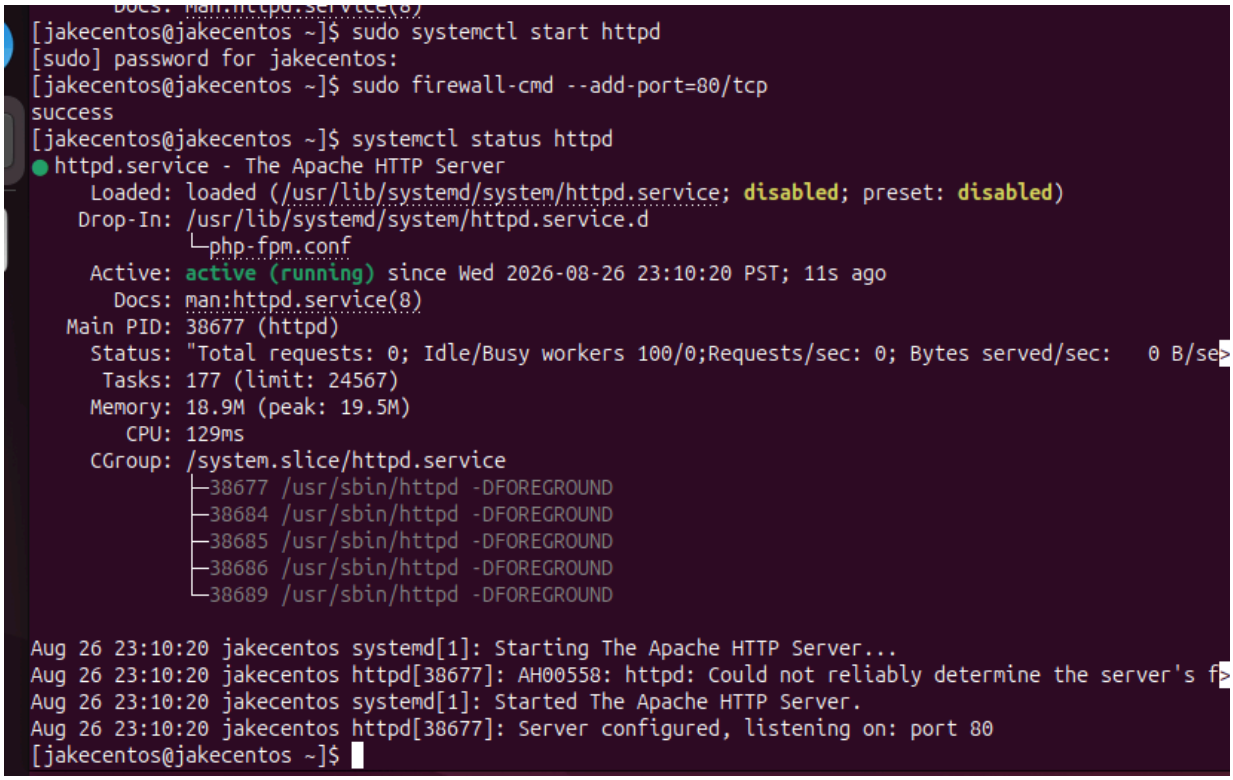


5.1 To activate, go to the CentOS VM terminal and enter the following:
systemctl status httpd

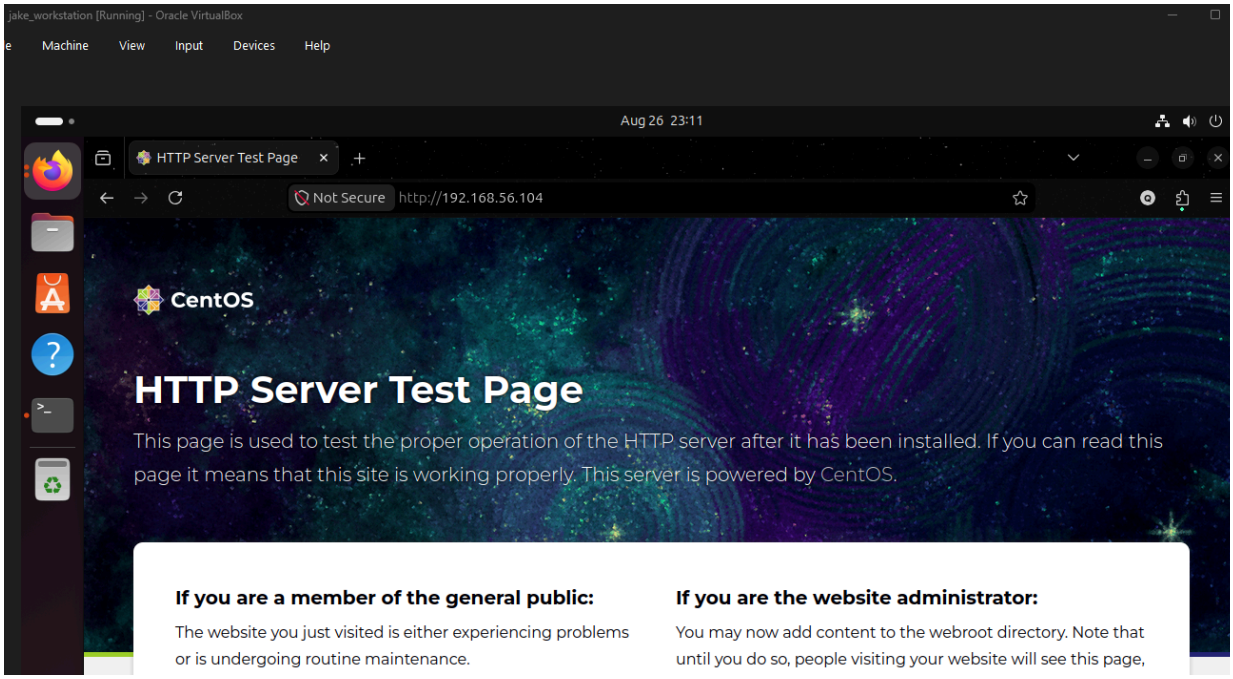
```
jakeworkstation@WS:~/ansible-lab$ ssh jakecentos@192.168.56.104
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Wed Aug 26 23:07:32 2026 from 192.168.56.101
[jakecentos@jakecentos ~]$ systemctl status httpd
○ httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
    Drop-In: /usr/lib/systemd/system/httpd.service.d
             └─php-fpm.conf
     Active: inactive (dead)
       Docs: man:httpd.service(8)
[jakecentos@jakecentos ~]$ A
```

The result of this command tells you that the service is inactive.
5.2 Issue the following command to start the service:
sudo systemctl start httpd
(When prompted, enter the sudo password)
sudo firewall-cmd --add-port=80/tcp
(The result should be a success)



5.3 To verify the service is already running, go to CentOS VM and type its IP address on the browser. Was it successful? (Screenshot the browser)



Task 2: Refactoring playbook

This time, we want to make sure that our playbook is efficient and that the codes are easier to read. This will also makes run ansible more quickly if it has to execute fewer tasks to do the same thing.

1. Edit the playbook install_apache.yml. Currently, we have three tasks targeting our Ubuntu machines and 3 tasks targeting our CentOS machine. Right now, we try to consolidate some tasks that are typically the same. For example, we can consolidate two plays that install packages. We can do that by creating a list of installation packages as shown below:

Make sure to save the file and exit.

Run ansible-playbook --ask-become-pass install_apache.yml and describe the Result.

jakeworkstation@WS: ~/ansible-lab

GNU nano 7.2 install_apache.yml

```
---
- hosts: all
  become: true

  tasks:

    - name: Install apache2 and php packages for Ubuntu
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
      when: ansible_distribution == "Ubuntu"

    - name: Install apache and php packages for CentOS
      dnf:
        name:
          - httpd
          - php
        state: latest
      when: ansible_distribution == "CentOS"
```

```
jakeworkstation@WS:~/ansible-lab$ nano install_apache.yml
jakeworkstation@WS:~/ansible-lab$ ansible-playbook -i inventory.yaml --ask-become-pass install_apache
.yml
BECOME password:

PLAY [all] *********************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [Install apache2 and php packages for Ubuntu] *****************************************************
skipping: [centos1]
ok: [ubuntu2]
ok: [ubuntu1]

TASK [Install apache and php packages for CentOS] ******************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

PLAY RECAP *********************************************************************************************
centos1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0
    ignored=0
ubuntu1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0
    ignored=0
ubuntu2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0
    ignored=0
```

2. Edit the playbook install_apache.yml again. In task 2.1, we consolidated the plays into one play. This time we can actually consolidated everything in just 2 plays. This can be done by removing the update repository play and putting the command update_cache: yes below the command state: latest. See below for reference:
Make sure to save the file and exit.
Run ansible-playbook --ask-become-pass install_apache.yml and describe the Result.

jakeworkstation@WS: ~/ansible-lab

GNU nano 7.2 install_apache.yml

```
---
- hosts: all
  become: true

  tasks:

    - name: Install apache2 and php packages for Ubuntu
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: Install apache and php packages for CentOS
      dnf:
        name:
          - httpd
          - php
        state: latest
        update_cache: yes
      when: ansible_distribution == "CentOS"
```

```
jakeworkstation@WS:~/ansible-lab$ ansible-playbook -i inventory.yaml --ask-become-pass install_apache
.yml
BECOME password:

PLAY [all] *********************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [Install apache2 and php packages for Ubuntu] *****************************************************
skipping: [centos1]
ok: [ubuntu1]
ok: [ubuntu2]

TASK [Install apache and php packages for CentOS] ******************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

PLAY RECAP *********************************************************************************************
centos1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0
    ignored=0
ubuntu1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0
    ignored=0
ubuntu2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0
    ignored=0

jakeworkstation@WS:~/ansible-lab$
```

3. Finally, we can consolidate these 2 plays in just 1 play. This can be done by declaring variables that will represent the packages that we want to install. Basically, the apache_package and php_package are variables. The names are arbitrary, which means we can choose different names. We also take out the line when: ansible_distribution. Edit the playbook install_apache.yml again and make sure to follow the below image. Make sure to save the file and exit.
Run ansible-playbook --ask-become-pass install_apache.yml and describe the Result.

```
jakeworkstation@WS: ~/ansible-lab
  GNU nano 7.2                                  install_apache.yml *
---
- hosts: all
  become: true

  vars:
    apache_package: apache2
    php_package: libapache2-mod-php

  tasks:

    - name: Install apache and php
      apt:
        name:
          - "{{ apache_package }}"
          - "{{ php_package }}"
        state: latest
        update_cache: yes
```

```
jakeworkstation@WS:~/ansible-lab$ ansible-playbook -i inventory.yaml --ask-become-pass install_apache
.yml
BECOME password:

PLAY [all] *********************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [Install apache and php] **************************************************************************
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
fatal: [centos1]: FAILED! => {"changed": false, "cmd": "apt-get update", "msg": "[Errno 2] No such fi
le or directory: b'apt-get'", "rc": 2, "stderr": "", "stderr_lines": [], "stdout": "", "stdout_lines"
: []}
ok: [ubuntu1]
ok: [ubuntu2]

PLAY RECAP *********************************************************************************************
centos1                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0
    ignored=0
ubuntu1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
    ignored=0
ubuntu2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
    ignored=0
```

4. Unfortunately, task 2.3 was not successful. It's because we need to change something in the inventory file so that the variables we declared will be in place.

Edit the inventory file and follow the below configuration:
Make sure to save the inventory file and exit.
Finally, we still have one more thing to change in our install_apache.yml file. In task 2.3, you may notice that the package is assign as apt, which will not run in CentOS. Replace the apt with package. Package is a module in ansible that is generic, which is going to use whatever package manager the underlying host or the target server uses. For Ubuntu it will automatically use apt, and for CentOS it will automatically use dnf. Make sure to save the file and exit. For more details about the ansible package, you may refer to this documentation:
ansible.builtin.package – Generic OS package manager — Ansible Documentation
Run ansible-playbook --ask-become-pass install_apache.yml and describe the result.

jakeworkstation@WS: ~/ansible-lab

GNU nano 7.2 inventory.yaml

```
all:
  children:

    ubuntu:
      hosts:
        ubuntu1:
          ansible_host: 192.168.56.105
          ansible_user: jakeserver
          os_type: Ubuntu

        ubuntu2:
          ansible_host: 192.168.56.103
          ansible_user: jakeserver2
          os_type: Ubuntu

      vars:
        apache_package: apache2
        php_package: libapache2-mod-php

    centos:
      hosts:
        centos1:
          ansible_host: 192.168.56.104
          ansible_user: jakecentos
          os_type: CentOS

      vars:
        apache_package: httpd
        php_package: php

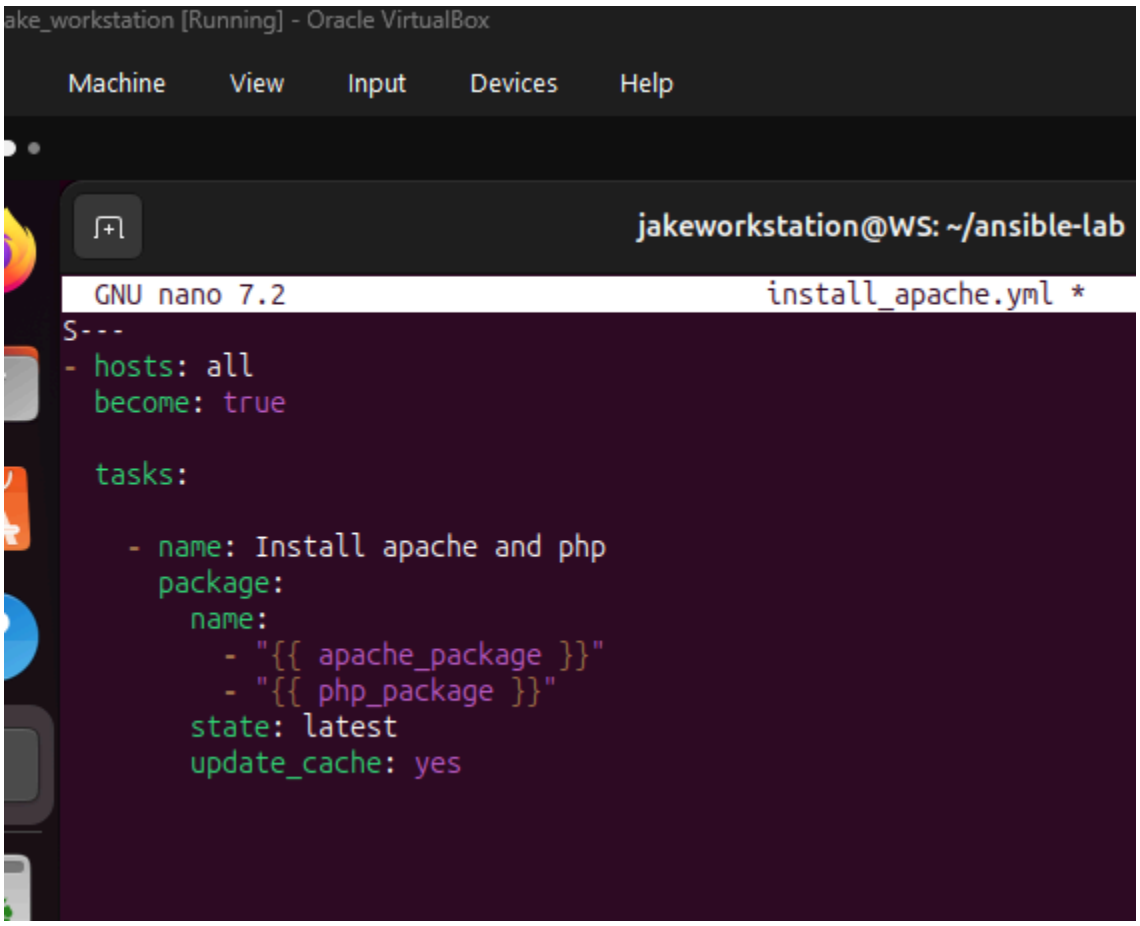
    linuxservers:
      children:
        ubuntu:
```

```
jakeworkstation@WS:~/ansible-lab$ ansible-inventory -i inventory.yaml --list
{
    "_meta": {
        "hostvars": {
            "centos1": {
                "ansible_host": "192.168.56.104",
                "ansible_user": "jakecentos",
                "apache_package": "httpd",
                "os_type": "CentOS",
                "php_package": "php"
            },
            "ubuntu1": {
                "ansible_host": "192.168.56.105",
                "ansible_user": "jakeserver",
                "apache_package": "apache2",
                "os_type": "Ubuntu",
                "php_package": "libapache2-mod-php"
            },
            "ubuntu2": {
                "ansible_host": "192.168.56.103",
                "ansible_user": "jakeserver2",
                "apache_package": "apache2",
                "os_type": "Ubuntu",
                "php_package": "libapache2-mod-php"
            }
        }
    },
    "all": {
        "children": [
            "ungrouped",
            "ubuntu",
            "centos",
            "linuxservers"
        ]
    },
    "centos": {
        "hosts": [
```

```
jakeworkstation@WS:~/ansible-lab$ ansible ubuntu -m debug -a "var=apache_package" -i inventory.yaml
ubuntu1 | SUCCESS => {
    "apache_package": "apache2"
}
ubuntu2 | SUCCESS => {
    "apache_package": "apache2"
}
jakeworkstation@WS:~/ansible-lab$
```

```
jakeworkstation@WS:~/ansible-lab$ ansible ubuntu -m debug -a "var=php_package" -i inventory.yaml
ubuntu1 | SUCCESS => {
    "php_package": "libapache2-mod-php"
}
ubuntu2 | SUCCESS => {
    "php_package": "libapache2-mod-php"
}
```

```
jakeworkstation@WS:~/ansible-lab$ ansible centos -m debug -a "var=php_package" -i inventory.yaml
centos1 | SUCCESS => {
    "php_package": "php"
}
```



```
jakeworkstation@WS:~/ansible-lab$ ansible-playbook -i inventory.yaml --ask-become-pass install_apache
.yml
BECOME password:

PLAY [all] *********************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [Install apache and php] **************************************************************************
ok: [ubuntu1]
ok: [ubuntu2]
ok: [centos1]

PLAY RECAP *********************************************************************************************
centos1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
    ignored=0
ubuntu1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
    ignored=0
ubuntu2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
    ignored=0
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Supplementary Activity:

1. Create a playbook that could do the previous tasks in Red Hat OS.

jakeworkstation@WS: ~/ansible-la

GNU nano 7.2 redhat_apache.yml *

```
---
- name: Configure Apache and PHP on Red Hat
  hosts: centos
  become: true

  tasks:

    - name: Install Apache and PHP
      package:
        name:
          - httpd
          - php
        state: latest

    - name: Start Apache
      service:
        name: httpd
        state: started
        enabled: true
```

```
jakeworkstation@WS:~/ansible-lab$ nano redhat_apache.yml
jakeworkstation@WS:~/ansible-lab$ ansible-playbook -i inventory.yaml --ask-become-pass redhat_apache.
yml
BECOME password:

PLAY [Configure Apache and PHP on Red Hat] *************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [centos1]

TASK [Install Apache and PHP] **************************************************************************
ok: [centos1]

TASK [Start Apache] ************************************************************************************
changed: [centos1]

PLAY RECAP *********************************************************************************************
centos1                    : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=0
    ignored=0

jakeworkstation@WS:~/ansible-lab$
```

Reflections:

Answer the following:

1. **Why do you think refactoring of playbook codes is important?**

- Refactoring makes the playbook shorter, cleaner, and easier to maintain. It also removes repeated code and allows the same playbook to work with different operating systems.

2. **When do we use the "when" command in playbook?**

- We use when when a task should only run under a specific condition. For example, we can use it to run an apt task only on Ubuntu and a different task only on CentOS.

Conclusion:

- This task really gave me an opportunity to use powerful Ansible ad hoc commands to perform system admin operations on the remote hosts with a privilege escalation. One more important thing I picked up was that the when statement enables you to run the commands as per operating system which means, for instance, that Ubuntu and CentOS will be using their own package managers. Refactoring playbooks taught me how to use package lists, variables, and the generic package module to shorten the repeat of tasks.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.